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BATCH: B1

Contents for Theory:

Data Visualization II

1. Use the inbuilt dataset 'titanic' as used in the above problem. Plot a box plot for distribution

of age with respect to each gender along with the information about whether they survived

or not. (Column names : 'sex' and 'age')

2. Write observations on the inference from the above statistics

```
In [1]: import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: df = sns.load_dataset('titanic')
```

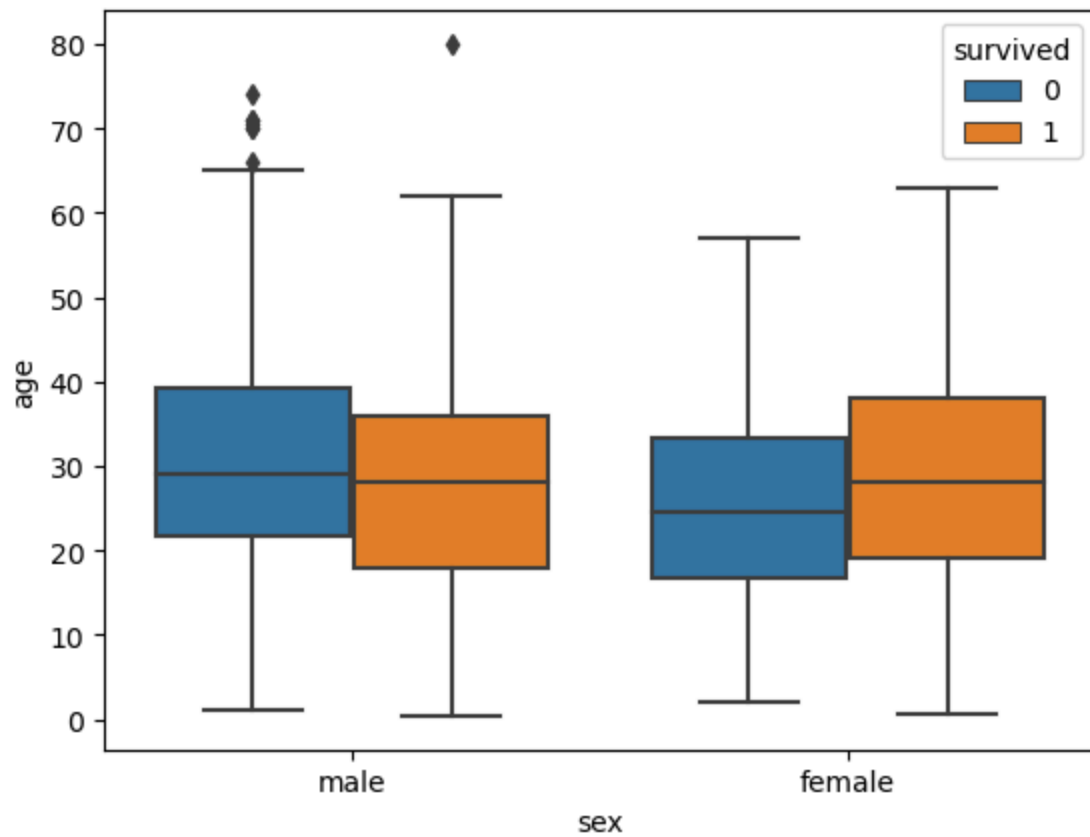
```
In [5]: print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

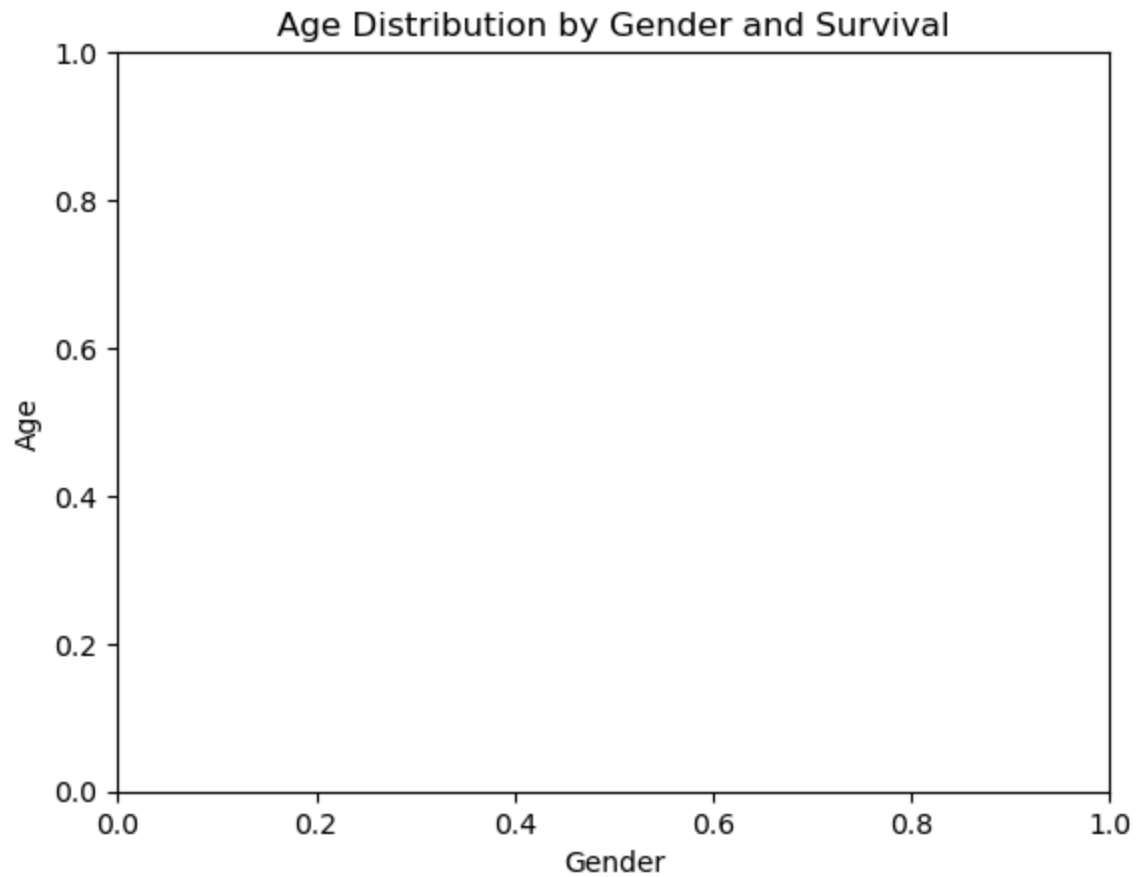
	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

```
In [6]: sns.boxplot(x='sex', y='age', hue='survived', data=df)
```

```
Out[6]: <Axes: xlabel='sex', ylabel='age'>
```



```
In [8]: plt.title("Age Distribution by Gender and Survival")  
plt.xlabel("Gender")  
plt.ylabel("Age")  
plt.show()
```



```
In [ ]:
```